

User Guide

Introduction

This user manual will provide installation instructions and key workflow steps for the two user types. Instructions for the admin user to log in and create a new major, including creating students and courses. The instructions for student users will cover the primary student workflow of viewing a term, viewing a course in the term, updating course status, disabling notifications, and adding notes to a course. Directions on dropping and adding a course will also be provided.

Installation and Using the Application

The application apk file is available for download on Github:

<https://ghammerle.github.io/WGU-D424-Capstone/>.

The source code is available on Gitlab:

[Task 3 · working · WGU GitLab Environment / Student Repos / ghamme3 / D424 Software Engineering Capstone · GitLab](#).

Installation (Visual Studio 2022 Setup)

All instructions below assume the computer is set up for mobile .NET MAUI development using Visual Studio 2022 Community Edition, git is installed, and the computer has internet access.

PC Setup for development and testing:

- Windows 11
- Intel i5-12400 12th Gen
- Radeon RX550/550 Series GPU
- 16GB of RAM

(Testing conducted on a PC with lesser graphics performance resulted in emulator launch issues. The emulator had to be started through the command line to avoid the graphics-related issue. “C:\Program Files (x86)\Android\android-sdk\emulator.exe” -avd pixel_7 -_api_52_1 -gpu swiftshader_indirect -verbose, it is best to avoid this complication.)

Open the Command Prompt. (Type cmd, and press Enter in the Windows search bar)

Create a working directory.

Navigate to the working directory and type git init to initialize git for this folder.

Navigate to the source code repository:

[Task 3 · working · WGU GitLab Environment / Student Repos / ghamme3 / D424 Software Engineering Capstone ·](#)

[GitLab](#)

Clone the repository, click code, and copy the path based on your account configuration.

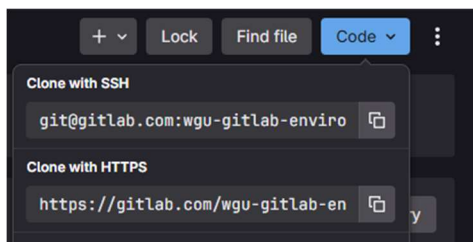


Figure 22 Git clone path

Return to Command Prompt, type git <path> (paste the copied path using Ctrl + V), and press Enter.

The repository will be cloned to the directory.

Check out the branch named working.

git checkout working

Open Visual Studio, open the solution C424 Capstone.sln

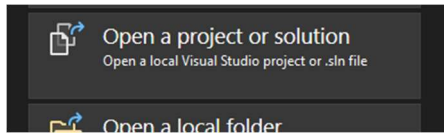


Figure 23 Open Solution

A directory navigation window will appear when the above selection is made.

Once the solution is loaded, use Tool → Android → Android Device Manager

Set up an Android device, the default was used to generate:

Name	OS	Processor	Memory	Resolution	Action
 Pixel 7 – API 35 + Google Play	Android 15.0 – API 35	x86_64	1 GB	1080 x 2400 420 dpi	 Stop

Figure 24 Android Pixel 7

Login

The application database comes seeded with three users.

- User Name: admin Password: Admin123!
- User Name: Beth.Santos Password: BethSantosPW
- User Name: Jake.Smith Password: JakeSmithPW

Students added to the database will have login credentials of

- User Name: First.Last Password: FirstLastPW

Students will be prompted to update the password on their initial login or if the admin resets it to the default password.

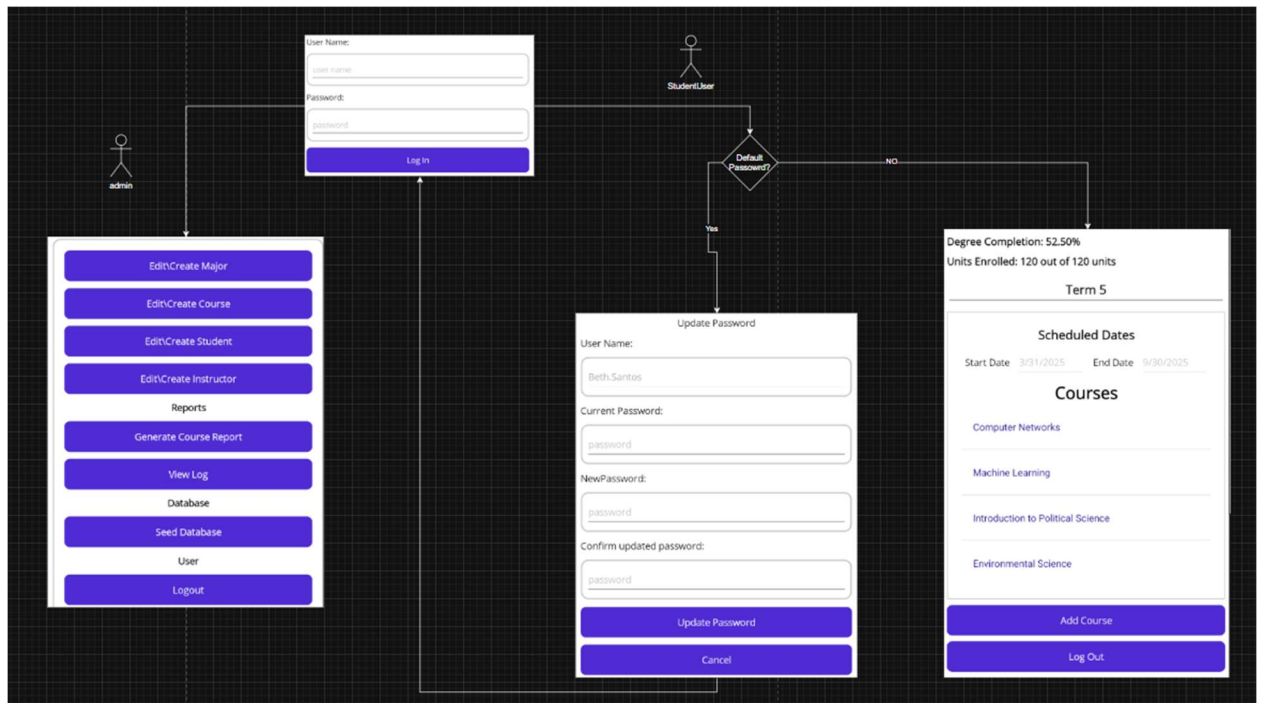


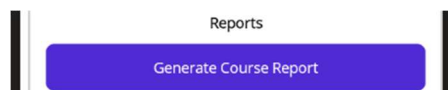
Figure 25 User Log in Flow

Administrator Workflow

The workflow detailed below will touch on major features available to the administrator user. Items covered will be adding and updating courses, majors, students, and instructors.

Viewing reports and database seeding will also be covered.

1. *Log in as admin.*
 - a. User Name: admin Password: Admin123!
2. *On the Select Task page, select Generate Course Report*



- a. *This will show a report of all courses in the database, the associated major, and the number of students enrolled. Notice: Test Major has zero students enrolled.*

b. Close the report

Test Major Course 15	Test Major	0
Test Major Course 16	Test Major	0
Test Major Course 17	Test Major	0
Test Major Course 18	Test Major	0

Close

3. Select Edit/Create Student



a. Using the Select Student Picker select a student.

i. Students who have started courses can only have their passwords reset to default.

ii. Cancel and return to the Edit Student page.

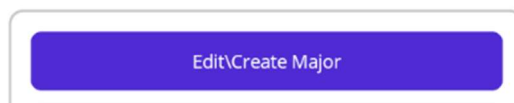
b. Using the select Major picker choose a major

Major:
Select Major

i. Notice only active majors are available. An active major is one with 20 associated courses.

c. Close the Edit Student Page using the Cancel button.

4. From the Select Task page select Edit\Create Major.



a. Select a Major

b. The associated course will appear.

Courses
Digital Media Literacy
Data Structures and Algorithms

c. Tap (click on) a course

Course Name: Data Structures and Algorithms

Select Major

Instructor
Lily Robinson

Instructor: Lily Robinson
Phone: 458-555-1217
Email: Lily.Robinson@this.edu

Course Duration: 2.5

Assessment One
Final Exam Objective

Assessment Two
Algorithm Implementation Project Performance

Update/Save Delete Cancel

i. If a course is assigned to a student only the instructor can be edited.

d. Click Cancel to return to the Edit Major page.

e. Click the Add Course Button.

Add Course

i. If the Major already has 20 courses, you cannot add a course.

f. From the Create/Edit Major page use the Major Picker to select Test Major.

i. This major was seeded with 18 courses for demonstration and training purposes.

g. Click Add Course

h. Enter a Course Name, select an instructor, and add at least one assessment.

Course Name: My New Course

Select Major

Instructor
Michael Davis

Instructor: Michael Davis
Phone: 458-555-1202
Email: Michael.Davis@this.edu

Course Duration: 2.5

Assessment One
Final Exam Performance

Assessment Two
Final Essay Objective

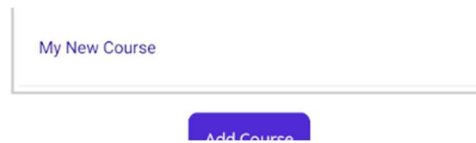
Update/Save Delete Cancel

i. *A course must have one assessment and no more than two.*

Assessments cannot be of the same type (Objective & Performance).

ii. *Click Update\Save to add the course to the major.*

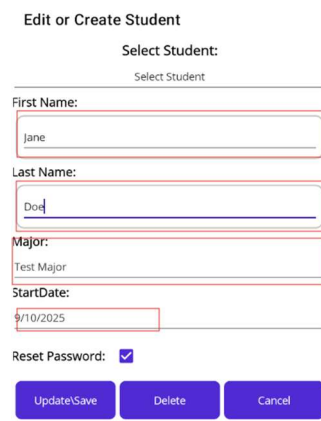
iii. *The course will appear in the list for the selected major.*



i. *Add a second course to Test Major; making it active.*

5. *Return to the Edit\Create student page.*

a. *Create a student and assign the major of Test Major. Provide first and last names and a start date. (Start Date can no be in the past.)*



b. *Click Update\Save to add the student.*

6. *To verify if a student was added to the major use the Generate Course Report button to view the course report.*

a. *Notice that Test Major courses have one student enrolled and some general education courses now have three students enrolled.*

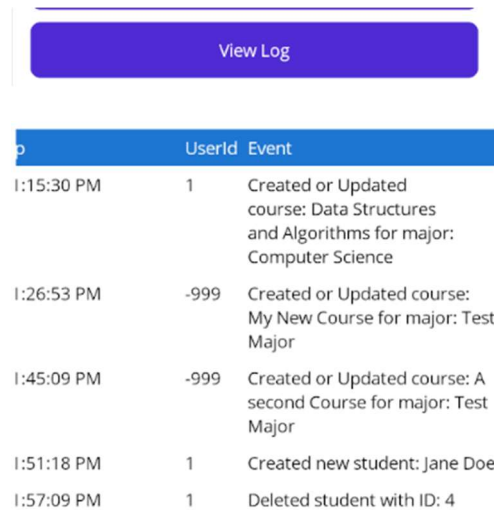
b. *The two courses added earlier will also appear in the report.*

7. *Delete a Major.*

- a. *Open the Create\Edit Major, select any major*
 - i. *Majors which have students enrolled in cannot be deleted.*
- b. *Return to the Task Selection page using Cancel.*

8. *Delete a student*

- a. *Open the Create\Edit Student page, select the newly added student.*
 - i. *Students who started courses cannot be deleted.*
- b. *Click the delete button to remove the selected student.*
- c. *Validate the student has been deleted using the Course Report.*
 - i. *The View Log button on the Select Task page will generate a report of the event log showing various events.*



The screenshot shows a blue button labeled 'View Log' at the top. Below it is a table with three columns: 'Time', 'UserId', and 'Event'. The table contains five rows of log entries.

Time	UserId	Event
1:15:30 PM	1	Created or Updated course: Data Structures and Algorithms for major: Computer Science
1:26:53 PM	-999	Created or Updated course: My New Course for major: Test Major
1:45:09 PM	-999	Created or Updated course: A second Course for major: Test Major
1:51:18 PM	1	Created new student: Jane Doe
1:57:09 PM	1	Deleted student with ID: 4

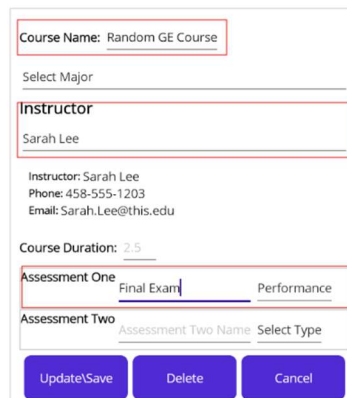
- d. *Open Create\Edit Major*
 - i. *Select Test Major from the Select Major picker.*
 - ii. *Click Delete, if no students are enrolled in this major the major can be deleted.*
- e. *Return to the Select Task Page and select Edit\Create Course.*

- i. *If a major does not contain 20 courses, it will be available from the Select Major picker.*



- ii. *Create a new course for General Education*

1. *Like adding course to a major directly, fill in the data but do not select a Major. Save the changes.*



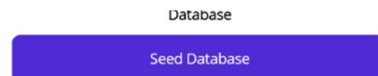
2. *Use Generate Course Report to validate the course was saved.*



Random GE Course	General Studies: 0
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- f. *Seeding the database*

- i. *The admin user can reset the database to the initial values using the Seed Database button on the Select Task page.*



- ii. *Click the button and wait for the dialog informing you that the database action is completed.*

Student Workflow

1. Log in as a student. (Beth Santos used for the example)

- a. There are two students provided.
 - i. Beth.Santos Password BethSantosPW
 - ii. Jake.Smith Passoword Jake.SmithPW
- b. Newly created students are defined with a pattern
 - i. FirstName.LastName Password : FirstNameLastNamePW
- c. When the password is the default students will be prompted to enter a new password.

Update Password

User Name:

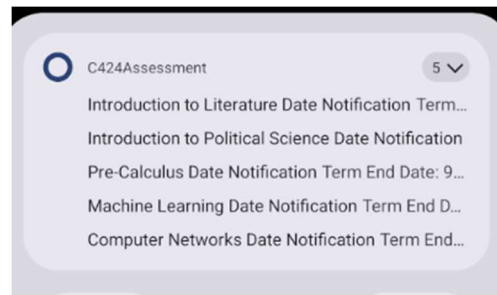
Current Password:

NewPassword:

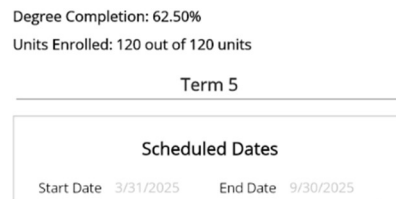
Confirm updated password:

- i. The current password must be entered, and the new password must be entered twice for confirmation. *(There is no secondary validation of passwords for length or reuse. This is done to simplify rapid testing.)*
- ii. After updating their password, users will need to log in using their new password.

- iii. The students provided were set up as if they were nearing the end of a term. When logged in, they should receive notifications.



- iv. Students will see their progress and Units enrolled. The application will open to their current term.



- d. Drop a course from a term.

- i. Select a Course

- 1. A course can be dropped from a term as long as the term will have at least 1 course remaining. If a general course is dropped, the user will need to add a course to get back to the required 120 units to graduate. Courses required for the enrolled major will move to the next available term.
 - 2. Pop-ups will occur to let the user understand the action.

Drop Course

This course is part of your degree plan. You will need to add another course to reach 120 units. Are you sure you want to drop it?

No Yes

Drop Course

This course is part of your major and will be moved to a later term.

Cancel Continue

3. Courses are randomly selected so that the examples may vary.

ii. When a course is selected, the course view will open

Course Editor

Introduction to Astronomy

Starting Date: 3/31/2025 Ending Date: 9/30/2025

Status: Planned Notifications: ☒

Instructor: David Martinez
Phone: 458-555-1204
Email: David.Martinez@this.edu

Notes

Performance Assessment

Star Chart Project Notifications: ☒

Due Date: 09/30/2025

Objective Assessment

Final Objective Exam Notifications: ☒

Due Date: 09/30/2025

Update

Remove Course

Return to Term

iii. Select Remove Course

1. After removing one general education course and one major course from the term, the user will see fewer courses, and the enrolled units will be updated.

Degree Completion: 55.00%

Units Enrolled: **117** out of 120 units

Term 5

Scheduled Dates

Start Date 3/31/2025 End Date 9/30/2025

Courses

Computer Networks

Introduction to Literature

Pre-Calculus

iv. Select Add Course

1. The user can select a course to add to the term.
2. The Select a Course picker will list courses available to the user.
 - a. This includes courses that may exist in future terms.
3. A search filter is available to filter the selection.

Course Selection Page

Search Selectable Courses:

Sci

Select a course.

Course to Add:

Select a course.

Physical Science

Environmental Science

Introduction to Political Science

Cancel

- v. Click Add Course once a selection has been made.

- vi. The course will be added to the current Term.

Degree Completion: 55.00%
Units Enrolled: 120 out of 120 units

Term 5

Scheduled Dates
Start Date: 3/31/2025 End Date: 9/30/2025

Courses

Computer Networks

Introduction to Literature

Pre-Calculus

Introduction to Political Science

Add Course

Log Out

- e. Updating Notifications

- i. Notifications are for a course and will be triggered if a course has not started and the term will end in the next 30 days.
- ii. Notifications can be disabled by course and assessment. Course settings take precedence.

Status: Planned Notifications: ☒

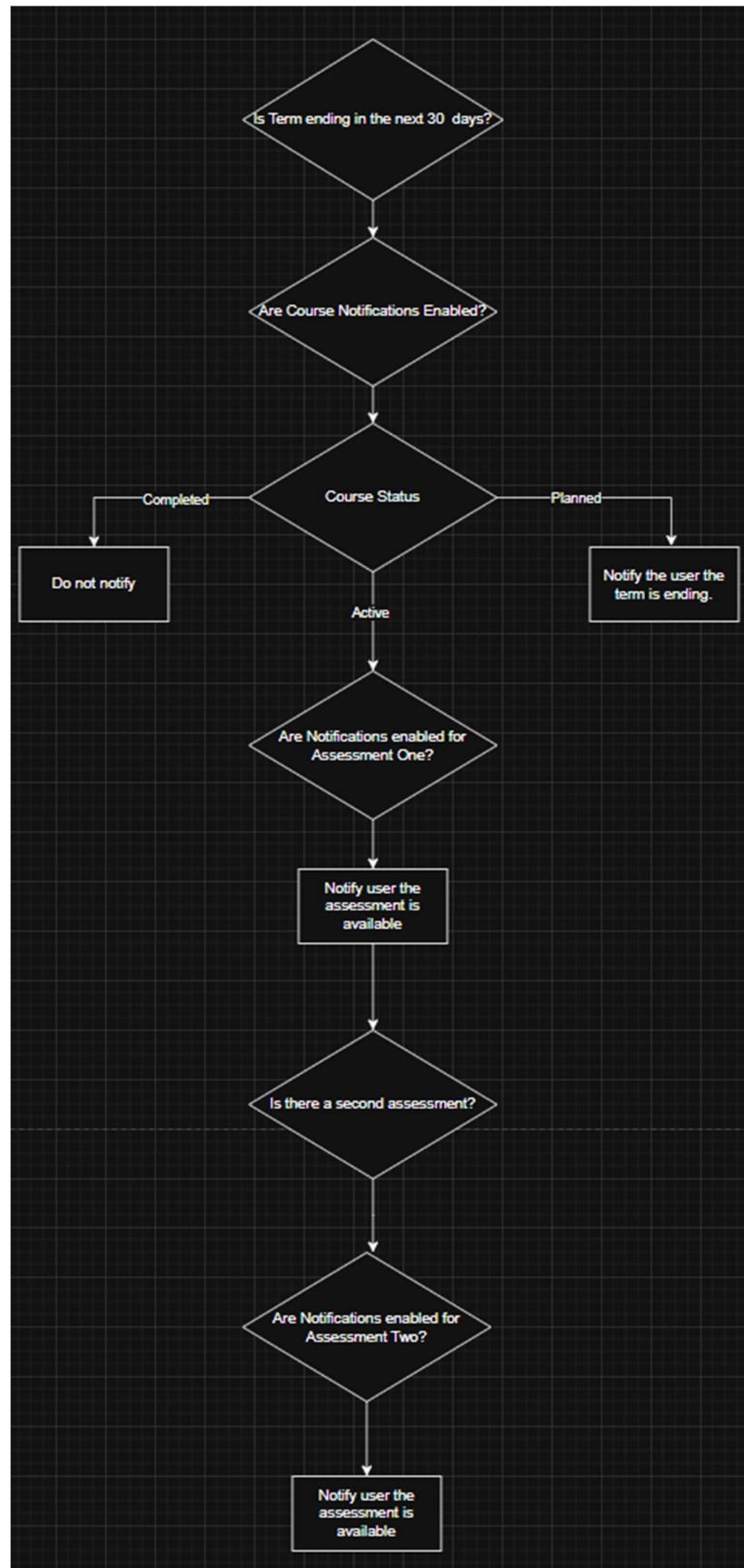
Instructor: Andrew Harris
Phone: 458-555-1212
Email: Andrew.Harris@this.edu

Notes

Performance Assessment
Policy Review Notifications: ☒
Due Date: 09/30/2025

Objective Assessment
Comparative Politics Quiz Notifications: ☒
Due Date: 09/30/2025

- iii. Notification flow chart



f. Updating Course Status

- i. Navigate to the current term
- ii. Use the status picker to set a course to completed.

Introduction to Political Science

Starting Date: 3/31/2025 Ending Date: 9/30/2025

Status: Completed Notifications: ☒

Instructor: Andrew Harris

- iii. Click Update to save the changes.
- iv. Notice that the degree completion percentage has been updated.

Degree Completion: 57.50%

Units Enrolled: 120 out of 120 units

g. Adding and sharing course notes.

- i. From the Term Page, select a course.
- ii. In the Course Page, click the Notes button.

Email: Andrew.Harris@this.edu

Notes

- iii. Add content in the note entry field.

Notes for Introduction to Political Scien...

Enter your note here.

Share

Update/Save Delete Cancel

- iv. Click Update/Save
- v. Return to the recently created note.

- vi. Use the Share button to bring up the sharing window.

